

CAT 2025

MBA FASTRACK



Lecture - 6

ALGEBRA

Functions

By- RAHUL BATHLA



TOPICS

to be covered

- **1** Functions – Basics
- **2** Composite Functions
- **3** Special Functions



Topics : Functions – Basics



WHAT IS A FUNCTION ?

A) $a + 2$
B) $x^2 + 2x$ ✓

$$f(a) = ax^2 + 2x$$
$$f(1) = 1(x^2) + 2x$$
$$= x^2 + 2x$$



A function is defined as a **relation between a set of inputs having one output each**. In simple words, a function is a relationship between inputs where each input is related to exactly one output.



VARIABLE

$$x = 2$$

$$f(x) = x^2 + 2x$$

$$f(2) = 2^2 + 2(2) = 4 + 4 = 8$$

$$f(2) = 8$$

↑ INPUT

→ OUTPUT

INPUT

FUNCTION

OUTPUT

$$f(x) = \sqrt{x-3} + 2$$

$$x = 4$$

$$\begin{aligned} f(4) &= \sqrt{4-3} + 2 \\ &= \sqrt{1} + 2 \end{aligned}$$

$$= 3$$

$$x = 1$$

$$\begin{aligned} f(1) &= \sqrt{1-3} + 2 \\ &= \sqrt{-2} + 2 \\ &\quad ?? \end{aligned}$$

??

QUESTION-1



#Q. If $f(x) = 2x^4 - 3x^{21} + x - 1$ and $g(x) = x(3x^{20} - 1 + 2x^3) + 3$. Find $f(2) + g(2)$.

A

66 ✓✓

B

88

C

100

D

124

$$f(x) = 2x^4 - 3x^{21} + x - 1$$

$$g(x) = 3x^{21} - x + 2x^4 + 3$$

$$f(2) + g(2) = 4(2)^4 + 2$$

$$= 64 + 2$$

$$= 66$$

QUESTION- 2



#Q.

If $f(x) =$

$$\boxed{2x + 3} \checkmark$$

$$\boxed{4x} \checkmark$$

$$2 \checkmark$$

$$x < -3 \quad \textcircled{1}$$

$$-3 \leq x < 2 \quad \textcircled{2} \quad \text{Find } f(-5) + f(1) + f(2).$$

$$2 \leq x \quad \textcircled{3}$$

INPUT		
$2(-5) + 3$	$4(1)$	2
-7	4	

$$\boxed{-1} \checkmark$$

A $-1 \checkmark$

B 5

C 4

D 1



Domain of Functions

$$f(x) = \sqrt{\underbrace{x+3}_{\rightarrow \text{positive}}}$$

$$f(-4) = ??$$

$$\text{DOMAIN} = x \in [-3, \infty)$$

$$x+3 \geq 0$$

$$\boxed{x \geq -3} \checkmark$$

The domain of a function is **the set of values that we are allowed to plug into our function.**



Square Root

$$f(x) = \sqrt{\text{poly.}}$$

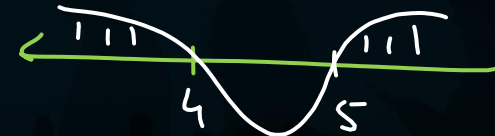
DOMAIN $\rightarrow$ $\boxed{\text{poly} \geq 0}$

QUAN. INSIDE SQ ROOT
 ≥ 0

Example

$$f(x) = \sqrt{x^2 - 9x + 20}$$

$$x^2 - 9x + 20 \geq 0$$
$$(x-4)(x-5) \geq 0$$



$$(-\infty, 4] \cup [5, \infty)$$

The domain of a function is **the set of values that we are allowed to plug into our function.**



Fractional Expressions

$$f(x) = \frac{N}{D}$$

$\downarrow$
0

Deno $\neq 0$

$$f(x) = \log_B A$$

$$A > 0 \checkmark$$

$$B > 0 \checkmark$$

$$B \neq 1 \checkmark$$

Example

$$f(x) = \frac{2x}{x+1}$$

$$x+1 \neq 0$$

$$x \neq -1$$

$$\text{DOMAIN} = \mathbb{R} - \{-1\}$$

QUESTION- 3



#Q. Find the domain of following function:

$$f(x) = \frac{1}{x^2 - 6x + 8} + \sqrt{x + 4} + \log_x(x + 3)$$

$$x^2 - 6x + 8 \neq 0$$

$$(x-4)(x-2) \neq 0$$

$$\boxed{\begin{matrix} x \neq 2 \\ x \neq 4 \end{matrix}}$$

$$x + 4 \geq 0$$

$$x \geq -4$$

$$\boxed{x \in [-4, \infty)}$$

$$x + 3 > 0$$

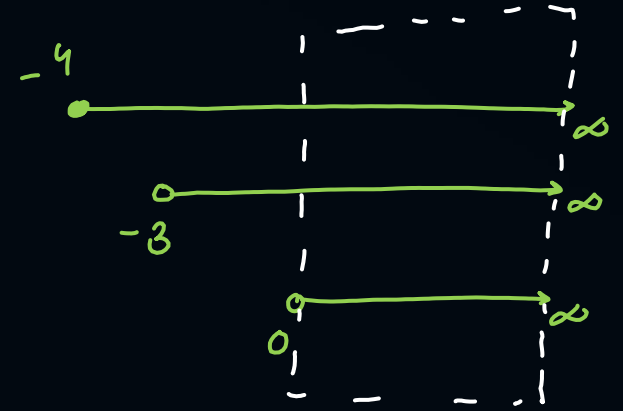
$$x > -3$$

$$\boxed{x \in (-3, \infty)}$$

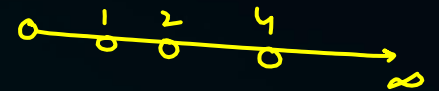
$$x > 0$$

$$\boxed{x \in (0, \infty)}$$

$$\boxed{x \neq 1}$$



$$x \in (0, \infty) - \{1, 2, 4\}$$



$$f(x) = \sqrt{\sqrt{x-3} - \sqrt{7-x}}$$

DOMAIN

$$\begin{aligned} x-3 &\geq 0 & 7-x &\geq 0 \\ x &\geq 3 & 7 &\geq x \quad \text{or} \quad x \leq 7 \\ x &\in [3, \infty) & x &\in (-\infty, 7] \end{aligned}$$

$$\sqrt{x-3} - \sqrt{7-x} \geq 0$$

$$\sqrt{x-3} \geq \sqrt{7-x}$$

$$x-3 \geq 7-x$$

$$2x \geq 10$$

$$x \geq 5$$

$$x \in [5, \infty)$$



Arithmetic of Functions



$$x \in [5, 7]$$



If $f(x) = x^2$ and $g(x) = 2x + 1$

$$(f + g)(2)$$

$$x^2 + 2x + 1$$

$$(x + 1)^2$$

$$(2 + 1)^2$$

$$3^2 = 9$$

$$f - g$$

$$x^2 - 2x - 1$$

$$f \cdot g$$

$$x^2(2x + 1)$$

$$f / g$$

$$\frac{x^2}{2x + 1}$$

$$\text{DOMAIN} = x \in \mathbb{R} - \left\{-\frac{1}{2}\right\}$$

QUESTION- 4 – Smart Approach



#Q. If $f(x) = x^2 - 7x$, and $g(x) = 7x - x^2 + 2$, then find

$$\underbrace{f(\sqrt{1}) + f(\sqrt{2}) + f(\sqrt{3}) \dots \dots f(\sqrt{10}) + g(\sqrt{1}) + g(\sqrt{2}) + g(\sqrt{3}) \dots \dots g(\sqrt{10})}$$

A 20 ✓

B 15

C 12

D 16

$$\left. \begin{array}{l} (f+g)(\sqrt{1}) = 2 \\ (f+g)(\sqrt{2}) = 2 \\ \vdots \\ (f+g)(\sqrt{10}) = 2 \end{array} \right\}$$

$$2 \times 10 = 20$$

$$(f+g)(x) = 2$$



Topics : Composite Functions

PYQ

2017 - 2019 ✓

$$f(x) = x^2 + 2x$$

$$g(x) = x + 1$$

$$f(\underbrace{g(x)}_{\text{INPUT}}) = ??$$

$$f \circ g(x) \rightarrow f \text{ of } g(x)$$

$$g \circ f(x) \rightarrow g \text{ of } f(x)$$

$$g[f(x)]$$

$$\text{Function} \rightarrow g \circ g(x) \text{ input}$$

INPUT 2

$$f(x) = x^2 + 2x$$

$$\downarrow$$

$$f(2) = 2^2 + 2(2)$$

$$= 8$$

$$g(x) = x + 1$$

$$g(f(x)) = f(x) + 1$$

$$= x^2 + 2x + 1$$

$$g(f(x)) = (x+1)^2$$

INPUT $g(x)$

$$f(x) = x^2 + 2x$$

$$\downarrow$$

$$f(\underbrace{g(x)}_{\text{INPUT}}) = [g(x)]^2 + 2g(x)$$

$$= (x+1)^2 + 2(x+1)$$

$$= (x+1)(x+1+2)$$

$$f(g(x)) = (x+1)(x+3)$$

QUESTION- 5



#Q. If $f(x) = \sqrt{x - 19}$, and $g(x) = x^2$ Find $f(\underbrace{g(10)}_{100})$.

A

9

$$\begin{aligned} f(100) &= \sqrt{100 - 19} \\ &= \sqrt{81} \\ &= 9 \end{aligned}$$

$$\begin{aligned} g(10) &= 10^2 \\ &= 100 \end{aligned}$$

$$\begin{aligned} f(100) \\ &= 9 \end{aligned}$$

B

10

C

-9

D

± 9

LINEAR

$$\begin{aligned} y &= \sqrt{81} \\ y &= 9 \end{aligned}$$

QUAD.

$$\begin{aligned} y^2 &= 81 \\ y &= \pm 9 \end{aligned}$$



#Q. If $f(x) = x^2 - 7x$, and $g(x) = x + 3$, then the minimum value of $f(g(x)) - 3x$.

A -20

B -15

C -12

D -16 ✓✓

$$f(x) = x^2 - 7x$$

$$f(g(x)) = [g(x)]^2 - 7(g(x))$$

$$= (x+3)^2 - 7(x+3)$$

$$= x^2 + 6x + 9 - 7x - 21$$

$$= x^2 - x - 12$$

$$\begin{aligned} f(g(x)) - 3x &= x^2 - x - 12 - 3x \\ &= x^2 - 4x - 12 \end{aligned}$$

$$x_{\min} = \frac{4}{2} = 2$$

$$4 - 8 - 12$$

$$4 - 20 = -16$$

QUESTION - 7 - CAT 2017 Slot 2



#Q. Let $f(x) = x^2$ and $g(x) = 2^x$, for all real x . Then the value of $f[f(g(x)) + g(f(x))]$ at $x = 1$ is

$$\begin{array}{l|l} g(x) = 2^x & f(x) = x^2 \\ g(1) = 2^1 & f(1) = 1^2 \\ = 2 & = 1 \end{array}$$

$$\begin{aligned} & f[f(g(1)) + g(f(1))] \\ & f[f(2) + g(1)] \\ & f[4 + 2] \end{aligned}$$

$$f[6] = 6^2 = 36$$

A

16

B

18

C

36 ✓

D

40



Topics : Functions – Special Types

TYPE 1 Question – Quadratic Polynomial



$F(x)$ is QUADRATIC
POLYNOMIAL

$$f(3) = 0 ; f(7) = 0$$

$$f(2) = -8$$

$$f(x) = k(x-3)(x-7)$$

$$f(2) = k(2-3)(2-7)$$

$$-8 = k(-1)(-5)$$

$$k = -\frac{8}{5}$$

$$f(x) = -\frac{8}{5}(x-3)(x-7)$$

$$f(8) = -\frac{8}{5}(5)(1)$$

$$\boxed{f(8) = -8}$$

Find $f(8)$

Question – 8



#Q. Let $f(x)$ be a quadratic polynomial in x for all real numbers x . If $f(2)=0$, $f(4)=0$, and $f(3) = -1$, then find $f(5)$.

A

2

B

3 ✓

C

1

D

4

$$f(x) = k(x-2)(x-4)$$

$$f(3) = k(3-2)(3-4)$$

$$-1 = k(1)(-1)$$

$$\boxed{k=1}$$

$$f(x) = (x-2)(x-4)$$

$$f(5) = (5-2)(5-4)$$

$$= 3(1)$$

$$= 3 \checkmark$$

TYPE 1 Question – Quadratic Polynomial

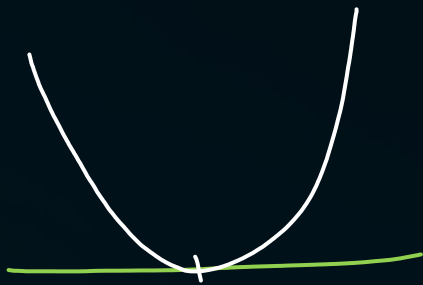


$$f(x) = k(x-a)^2$$

$F(x)$ is QUADRATIC
POLYNOMIAL

$$f(x) \geq 0$$

$$f(3) = 0$$



$$\begin{aligned} f(x) &= k(x-3)^2 \\ f(7) &= k(7-3)^2 \\ 64 &= k(4)^2 \end{aligned}$$

$$\begin{aligned} f(x) &= 4(x-3)^2 \\ f(9) &= 4(9-3)^2 \\ &= 144 \checkmark \end{aligned}$$

Find $f(9)$

$$f(7) = 64$$

#Q. Let $f(x)$ be a quadratic polynomial in x such that $f(x) \geq 0$ for all real numbers x . If $f(2)=0$ and $f(4)=6$, then $f(-2)$ is equal to

A 12

B 36

C 24 ✓

D 6

$$f(x) = k(x-2)^2$$

$$f(4) = k(4-2)^2$$

$$6 = 4k$$

$$k = \frac{3}{2}$$

$$f(x) = \frac{3}{2}(x-2)^2$$

$$f(-2) = \frac{3}{2}(-2-2)^2$$
$$= 24$$

#Q. If $f_1(x) = x^2 + 11x + n$ and $f_2(x) = x$, then the largest positive integer n for which the equation $\underline{f_1(x) = f_2(x)}$ has two distinct real roots, is:
(TITA)

$$x^2 + 11x + n = x$$

$$x^2 + 10x + n = 0 \begin{cases} \alpha \\ \beta \end{cases}$$

$$D > 0$$

$$100 - 4n > 0$$

$$\boxed{n = 24}$$

$$-4n > -100$$

$$4n < 100$$

$$n < 25$$

QUAD. func

$$(i) f(2)=0 \quad f(3)=0$$

$$f(x) = k(x-2)(x-3)$$

$$(ii) f(x) \geq 0 \quad f(2)=0$$

$$f(x) = k(x-2)^2$$

$$(iii) f(x) \geq 0$$

$$f(x) = k(x-a)^2$$

$$(iv) f(3)=0$$

$$f(x) = k(x-3)(x-a)$$

$$(v) f(x) \rightarrow \text{QUAD.}$$

$$f(x) = ax^2 + bx + c$$

$$f(x) \rightarrow \text{Linear func}$$

$$f(x) = ax + b$$

CUBIC func

$$f(1)=0 \quad f(5)=0$$

$$f(x) = k(x-1)(x-5)(x-a)$$

CUBIC func

$$f(x) = ax^3 + bx^2 + cx + d$$

TYPE 2 Question – $F(x)$ & $F(1/x)$



$$x \rightarrow \frac{1}{x}$$

$$2x \rightarrow \frac{2}{x}$$

$$\frac{1}{x} \rightarrow x$$

$$x^2 \rightarrow \frac{1}{x^2}$$

$$f(x) + 2f\left[\frac{1}{x}\right] = 3x$$

$$\boxed{6\text{sec}} \quad a + 2b = 3x$$

$$\boxed{2\text{sec}} \quad b + 2a = \frac{3}{x}$$

$$\rightarrow \boxed{f(x) + 2f\left[\frac{1}{x}\right] = 3x}$$

$$\underline{x \rightarrow \frac{1}{x}} \rightarrow \boxed{f\left(\frac{1}{x}\right) + 2f(x) = \frac{3}{x}}$$

$$\boxed{\checkmark f(x) = a \checkmark}$$

$$f\left(\frac{1}{x}\right) = b$$

$$a + 2b = 3x$$

$$2a + b = \frac{3}{x}$$

$$\times 2 \quad 4a + 2b = \frac{6}{x}$$

$$a + 2b = 3x$$

$$3a = \frac{6}{x} - 3x$$

$$f(x) = a = \frac{2}{x} - x$$

$$\boxed{f(x) = \frac{2}{x} - x}$$

QUESTION- 11



#Q. If $f(x) + 3f\left(\frac{1}{x}\right) = x$; Find $f(3)$.

A $-1/6$

B $-3/2$

C $1/4$

D $-1/4$ ✓

$$\begin{aligned} a + 3b &= x \\ 9a + 3b &= \frac{3}{x} \\ \hline -8a &= x - \frac{3}{x} \end{aligned}$$

$$\begin{aligned} a + 3b &= x \\ b + 3a &= \frac{1}{x} \end{aligned}$$

$$a = f(x) = -\frac{1}{8} \left(x - \frac{3}{x} \right)$$

$$= -\frac{1}{8} \left(3 - \frac{3}{3} \right)$$

$$= -\frac{1}{8} (2)$$

$$= -\frac{1}{4}$$

$$\begin{aligned} x &\rightarrow \frac{1}{x} \\ f(x) &\rightarrow f\left(\frac{1}{x}\right) \\ a &\rightarrow b \end{aligned}$$

$$\begin{aligned} \frac{1}{x} &\rightarrow x \\ f\left(\frac{1}{x}\right) &\rightarrow f(x) \\ b &\rightarrow a \end{aligned}$$

QUESTION- 12 (CAT 2024 – Slot 3)



#Q. For any non-zero real number x , let $f(x) + 2f\left(\frac{1}{x}\right) = 3x$. Then, the sum of all the possible values of x for which $f(x) = 3$.

A 2

B -3 ✓

C -2

D 3

$$\begin{aligned} a + 2b &= 3x \\ 4a + 2b &= \frac{6}{x} \\ \hline -3a &= 3x - \frac{6}{x} \end{aligned}$$

$$f(x) = \frac{2}{x} - x = 3$$

$$= 2 - x^2 = 3x$$

$$\boxed{x^2 + 3x - 2 = 0}$$

$$\text{Sum} = -3 \checkmark$$

$$f(x) = a^x$$

$$f(y) = a^y$$



INKO YAAD KARNA HAI:

COND.

Ass.

• If $f(x) \cdot f(y) = f(x + y)$; then $f(x) = a^x$

• If $f(x) + f(y) = f(x + y)$; then $f(x) = kx$

• If $f(x) \cdot f(y) = f(xy)$; then $f(x) = x^n$

• If $f(x) \cdot f(1/x) = f(x) + f(1/x)$; then $f(x) = 1 \pm x^n$

$$\text{LHS} = f(x) \cdot f(y)$$

$$= a^x \cdot a^y$$

$$= a^{x+y}$$

$$\text{RHS} \quad f(x+y) = a^{x+y}$$

QUESTION – 13 (CAT 2020 Slot 3)



#Q. If $f(x) \cdot f(y) = f(x + y)$; and $f(5) = 4$, then $f(10) - f(-10) =$

A

0

B

15.9375

C

3

D

14.0625

$$f(x) = a^x$$

$$f(5) = a^5$$

$$4 = a^5$$

$$16 = a^{10}$$

$$x=5 \quad y=5$$

$$f(5) \cdot f(5) = f(5+5)$$

$$(4)(4) = f(10) = 16 \checkmark$$

$$f(10) = a^{10} = 16$$

$$f(-10) = a^{-10} = \frac{1}{a^{10}} = \frac{1}{16}$$

$$16 - \frac{1}{16}$$

QUESTION - 14



#Q. If $f(x) \cdot f(1/x) = f(x) + f(1/x)$; and $\underline{f(3) = 10}$, Find $f(5)$.

A 21

B 25

C 26 ✓

D 24

$$f(x) = 1 \pm x^n$$

$$f(3) = 1 \pm 3^n$$

$$10 = 1 \pm 3^n$$

$$10 = 1 + 3^n$$

$$9 = 3^n$$

$$\boxed{n=2} \checkmark$$

$$f(x) = 1 + x^2$$

$$\begin{aligned} f(5) &= 1 + 5^2 \\ &= 1 + 25 \\ &= 26 \end{aligned}$$



Topics : Functions Special Type – 4 : Series

OBSERVE KARNA HAI INKO:



Difference Series

- $f(x) - f(x-1) = 4$
- $f(x) - f(x-1) = x$

1 + 2 ... 100 = 5050
2021-SLOT 3

$$\begin{cases} f(1) = 4 \\ f(10) = ?? \end{cases}$$

$$\cancel{f(11) - f(10) = 4}$$

$$\checkmark f(10) - f(9) = 4$$

$$\cancel{f(9) - f(8) = 4}$$

$$\cancel{f(8) - f(7) = 4}$$

⋮

⋮

$$\cancel{f(2) - f(1) = 4}$$

$$f(10) - f(1) = 36$$

$$\begin{aligned} f(10) &= 36 + 4 \\ &= 40 \checkmark \end{aligned}$$

$$f(2) - f(1) = 4$$

$$f(3) - f(2) = 4$$

$$\cancel{f(10) - f(9) = 10}$$

$$\cancel{f(9) - f(8) = 9}$$

⋮

⋮

⋮

⋮

$$\cancel{f(2) - f(1) = 2}$$

$$f(10) - f(1) = 2 + 3 + 4 + \dots + 10$$

$$f(10) - f(1) = 54$$

$$f(10) = 58 \checkmark$$

QUESTION - 15



#Q. $f(x) = f(x - 1) + x$, and $f(1) = 6$, find $f(10)$.

HW

A 20

B 30

C 50

D 60

$$f(x) - f(x-1) = x$$

QUESTION - 16



#Q. $f(x) - f(x - 1) = f(x)f(x - 1)$, and $f(1) = 0.1$, find $f(10)$.

A 1

B 0.1

HW

C 2

D 0.2

Modulus

MOD KA GOD - MBA WALLAH

3Hours



THANK
You

